

AP CSP | Lesson 19: Procedures & Libraries

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: A

A follows zero elements. B invents a visit. C incorrectly retains a previous input. D assumes indexing not required by FOR EACH.

Q2. Answer: C

C is 150/60. A returns the input; B truncates decimal division; D subtracts 60 without instruction.

Q3. Answer: B

B binds arguments by position. A confuses values with names. C reverses order. D confuses procedure and parameter names.

Q4. Answer: D

D distinguishes the effect from a returned result. A invents a return contract. B confuses procedure flow with program termination. C ignores DISPLAY.

Q5. Answer: A and D

A supports generalization. D describes useful API documentation. B ignores the need to understand behavior. C confuses defining with calling.

Q6. Answer: B

B reflects immediate return from the current call. A and C execute unreachable code. D incorrectly ends the entire caller, which can continue.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 0.75; (b) 2; (c) true; (d) false.

1 mark: 0.75 hours.

1 mark: 2 hours.

1 mark: true at the boundary.

1 mark: false below the goal.

Q8. 4 marks

Answer: PROCEDURE toHours(minutes)

```
{  
  RETURN(minutes / 60)  
}  
hours ← toHours(90)
```

1 mark: Correct named procedure definition.

1 mark: minutes is the input parameter used in the calculation.

1 mark: Returns minutes/60 rather than only displaying it.

1 mark: Calls toHours(90) and assigns its result to hours.

Q9. 4 marks

Answer: PROCEDURE reachedGoal(minutes, goal)

```
{  
  RETURN(minutes ≥ goal)  
}
```

met ← reachedGoal(20, 30)

1 mark: Defines both parameters in the named procedure.

1 mark: Compares minutes ≥ goal inclusively using the parameters.

1 mark: Returns the Boolean result.

1 mark: Correct two-argument call assigned to met.

Q10. 4 marks

Answer: (a) 8. (b) No. (c) RETURN immediately transfers control back to the caller. (d) 10.

1 mark: Returned value 8.

1 mark: 99 is not displayed.

1 mark: Explains immediate return from the current procedure call.

1 mark: Caller stores 10.

Q11. 4 marks

Answer: (a) Minutes. (b) Seconds. (c) 150. (d) Yes: use DISPLAY(seconds(2.5)) or display a stored returned result.

1 mark: Input unit minutes.

1 mark: Returned unit seconds.

1 mark: Correct conversion 150.

1 mark: A separate caller DISPLAY is required to show the value.

Q12. 4 marks

Answer: (a) reachedGoal. (b) minutes and goal. (c) Shared comparison logic can be maintained in one body instead of three copies. (d) Preserve the documented inputs, returned meaning and other promised behavior.

1 mark: Meaningful helper name.

1 mark: Two parameters representing observed minutes and target.

1 mark: Concrete reduction of duplicated logic or one-place maintenance.

1 mark: Requires preserving the caller-visible contract, not arbitrary changes.